VTPEH 6207 - CheckPoint 2

Ke Wu

2026-02-06

3.1 Background Chronic diseases such as obesity, diabetes, and cardiovascular disease act as major public health concerns in New York State. Behavioral risk factors like physical inactivity, poor diet, and smoking significantly contribute to these concerns. Analyzing BRFSS data helps explore how health behaviors relate to chronic disease risk and identify vulnerable populations. The motivation of this project is driven by a better understanding on behavioral factors to support more efficient interventions and decisions that improve community health outcomes.

Project Question What is the relationship between selected behavioral risk factors and chronic disease indicators in adults, in New York State based on 2023 BRFSS survey data?

Data Overview This dataset comes from the New York State Department of Health (NYSDOH) Behavioral Risk Factor Surveillance System (BRFSS) 2023 survey. It is a telephone survey coordinated nationally by the CDC and administered at the state level. The study population includes non-institutionalized adults aged 18 years and older living in New York State. The data represent self-reported health behaviors, chronic disease status, and preventive health practices. The data collection period is 2023. #loading data

```r
# Packages
library("janitor")
```

```
##
## Attaching package: 'janitor'

## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
# Load file
data_BRFSS=read.csv("/Users/kewu/Desktop/NYSDOH_BRFSS_SurveyData_2023.csv")

# Preview the data
head(data_BRFSS)
names(data_BRFSS)
str(data_BRFSS)
summary(data_BRFSS)
```

3.2.1

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
var_table <- data.frame(
  Variable = names(data_BRFSS),
  Class = sapply(data_BRFSS, class),
  Type = sapply(data_BRFSS, function(x){
    if(is.numeric(x) | is.integer(x)) "Continuous"
    else if(is.factor(x) | is.character(x)) "Categorical"
    else "Other"
  })
)

# <=10 variable
vars_keep <- c("SEXVAR", "AGE", "_BMI5", "_SMOKER3", "DIABETE4",
               "GENHLTH", "EXERANY2", "INCOME3")

var_table_10 <- var_table %>% filter(Variable %in% vars_keep)

knitr::kable(var_table_10,
             caption = "Table 1. Overview of selected variables from NYSDOH BRFSS 2023.")
```

Table 1: Table 1. Overview of selected variables from NYSDOH BRFSS 2023.

	Variable	Class	Type
SEXVAR	SEXVAR	character	Categorical
GENHLTH	GENHLTH	character	Categorical
EXERANY2	EXERANY2	character	Categorical
DIABETE4	DIABETE4	character	Categorical
AGE	AGE	character	Categorical
INCOME3	INCOME3	character	Categorical

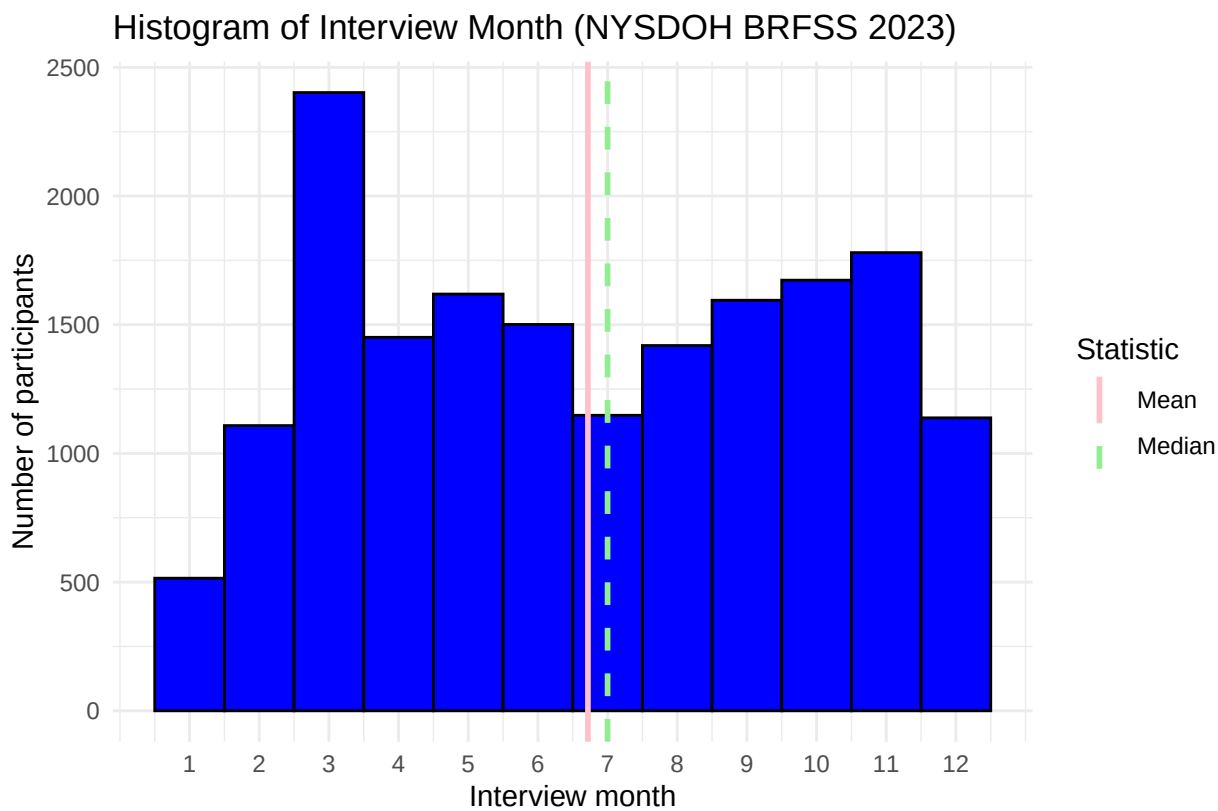
3.2.2.1(IMONTH)

```r
library(ggplot2)

x <- data_BRFSS$IMONTH

#mean and median
x_mean <- mean(x, na.rm = TRUE)
x_median <- median(x, na.rm = TRUE)
#mean/median lines and legend
ggplot(data_BRFSS, aes(x = IMONTH)) +
  geom_histogram(binwidth = 1, boundary = 0.5, color = "black", fill = "blue") +
  geom_vline(aes(xintercept = x_mean, color = "Mean"), linewidth = 1) +
```

```
  geom_vline(aes(xintercept = x_median, color = "Median"), linetype = "dashed", linewidth = 1) +
  scale_color_manual(values = c("Mean" = "pink", "Median" = "lightgreen")) +
  scale_x_continuous(breaks = 1:12) +
  labs(
    title = "Histogram of Interview Month (NYSDOH BRFSS 2023)",
    x = "Interview month",
    y = "Number of participants",
    color = "Statistic",
    caption = "Figure 1. Distribution of interview month in the NYSDOH BRFSS 2023 dataset. Solid red li
  ) +
  theme_minimal()
```



w month in the NYSDOH BRFSS 2023 dataset. Solid red line = mean; dashed blue line = median.

3.2.2.2

```
library(ggplot2)

data_BRFSS$month <- as.numeric(data_BRFSS$IMONTH)
data_BRFSS$day <- as.numeric(data_BRFSS$IDAY)


ggplot(data_BRFSS, aes(x = month, y = day)) +
  geom_point(alpha = 0.4, color = "pink") +

  scale_x_continuous(breaks = 1:12) +
  scale_y_continuous(breaks = seq(0, 31, 5)) +
```

```r
  labs(
    title = "Scatterplot of Interview Month vs Interview Day (BRFSS 2023)",
    x = "Interview Month (1 = Jan ... 12 = Dec)",
    y = "Interview Day of Month",
    caption = "Figure 2. Scatterplot showing distribution of interview timing by month and day in NYSDO
  ) +

  theme_minimal()
```

Figure 2. Scatterplot showing distribution of interview timing by month and day in NYSDOH BRFSS 2023 dataset.

4 This document was generated using CHATGTP to debugging the diagram code